

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 12
 - Limits of sequences

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$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } d(x_n, L) < \epsilon \quad \forall n \geq N.$$

We say $\{x_n\}$ **converges** to L and L is the **limit of the sequence**.

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This means $x_n = 1/n = L$ for all $n \geq N$, which is a contradiction!



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This is a contradiction.



Existence of limits

Theorem

If a sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$, then the range

$$X = \{x_1, x_2, \dots\} \subseteq M$$

is a bounded set and L is an adherent point of X .

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When the range of a sequence is bounded, we call the sequence $\{x_n\}$ bounded.

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If we take

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then $d(x_n, L) \leq R$ for all N .

Therefore $X \subseteq B_M(L, R)$ and in particular X is bounded. □

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Theorem

If S is a subset of a metric space (M, d) and $L \in M$ is an adherent point of S , then there exists a sequence $\{x_n\}$ of elements of S which converges to L .

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We claim $\lim x_n = L$.

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Since $\epsilon > 0$ was arbitrary, this proves convergence. □